11 CRUD

XML

```xml
<?xml version="1.0" encoding="utf-8"?>

<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="16dp">

    <EditText
        android:id="@+id/id"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="ID" />

    <EditText
        android:id="@+id/name"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Name" />

    <EditText
        android:id="@+id/age"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Age" />

    <Button
        android:id="@+id/add"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Add" />

    <Button
        android:id="@+id/view"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="View" />

    <Button
        android:id="@+id/update"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Update" />

    <Button
        android:id="@+id/delete"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Delete" />
```

```
    <TableLayout
        android:id="@+id/table"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:stretchColumns="*"/>
</LinearLayout>
```

JAVA

```
package com.example.crudwithtable;

import androidx.appcompat.app.AppCompatActivity;
import android.database.Cursor;
import android.os.Bundle;
import android.widget.*;

public class MainActivity extends AppCompatActivity {

    EditText id,name,age;
    DBHelper db;
    TableLayout table;

    protected void onCreate(Bundle b) {
        super.onCreate(b);
        setContentView(R.layout.activity_main);

        id=findViewById(R.id.id);
        name=findViewById(R.id.name);
        age=findViewById(R.id.age);
        table=findViewById(R.id.table);

        db=new DBHelper(this);

        findViewById(R.id.add).setOnClickListener(v->
                db.add(id.getText()+"",name.getText()+"",age.getText()+""));

        findViewById(R.id.update).setOnClickListener(v->
                db.update(id.getText()+"",name.getText()+"",age.getText()+""));

        findViewById(R.id.delete).setOnClickListener(v->
                db.delete(id.getText()+""));

        findViewById(R.id.view).setOnClickListener(v->show());
    }

    void show() {
        table.removeAllViews();
        Cursor c=db.view();
```

```
        TableRow h=new TableRow(this);
        h.addView(txt("ID"));
        h.addView(txt("Name"));
        h.addView(txt("Age"));
        table.addView(h);

        while(c.moveToNext()) {
            TableRow r=new TableRow(this);
            r.addView(txt(c.getString(0)));
            r.addView(txt(c.getString(1)));
            r.addView(txt(c.getString(2)));
            table.addView(r);
        }
    }

    TextView txt(String s) {
        TextView t=new TextView(this);
        t.setText(s);
        return t;
    }
}
```

DB

```
package com.example.crudwithtable;

import android.content.*;
import android.database.Cursor;
import android.database.sqlite.*;

public class DBHelper extends SQLiteOpenHelper {

    public DBHelper(Context c) {
        super(c,"db",null,1);
    }

    public void onCreate(SQLiteDatabase db) {
        db.execSQL("create table user(id integer primary key,name text,age text)");
    }

    public void onUpgrade(SQLiteDatabase db,int o,int n) {
        db.execSQL("drop table if exists user");
        onCreate(db);
    }

    void add(String i,String n,String a) {
        ContentValues cv=new ContentValues();
        cv.put("id",i); cv.put("name",n); cv.put("age",a);
        getWritableDatabase().insert("user",null,cv);
    }
```

```
    Cursor view() {
        return getReadableDatabase().rawQuery("select * from user",null);
    }

    void update(String i,String n,String a) {
        ContentValues cv=new ContentValues();
        cv.put("name",n); cv.put("age",a);
        getWritableDatabase().update("user",cv,"id=?",new String[]{i});
    }

    void delete(String i) {
        getWritableDatabase().delete("user","id=?",new String[]{i});
    }
```

12 DB LOGIN

XML

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="24dp"
    android:gravity="center">

    <EditText
        android:id="@+id/etUsername"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Username"/>

    <EditText
        android:id="@+id/etPassword"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Password"
        android:inputType="textPassword"
        android:layout_marginTop="10dp"/>

    <Button
        android:id="@+id/btnLogin"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Login"
        android:layout_marginTop="20dp"/>

    <Button
        android:id="@+id/btnRegister"
```

```
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Register"
        android:layout_marginTop="10dp"/>

</LinearLayout>
```

JAVA

```
package com.example.dblogin;

import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.Toast;

import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    EditText etUsername, etPassword;
    Button btnLogin, btnRegister;
    DBHelper db;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        etUsername = findViewById(R.id.etUsername);
        etPassword = findViewById(R.id.etPassword);
        btnLogin = findViewById(R.id.btnLogin);
        btnRegister = findViewById(R.id.btnRegister);

        db = new DBHelper(this);

        btnRegister.setOnClickListener(v -> {
            String user = etUsername.getText().toString();
            String pass = etPassword.getText().toString();

            if (user.isEmpty() || pass.isEmpty()) {
                Toast.makeText(this, "Fill all fields", Toast.LENGTH_SHORT).show();
            } else {
                Boolean insert = db.insertData(user, pass);
                if (insert)
                    Toast.makeText(this, "Registered Successfully", Toast.LENGTH_SHORT).show();
                else
                    Toast.makeText(this, "User already exists", Toast.LENGTH_SHORT).show();
            }
        });
```

```
        btnLogin.setOnClickListener(v -> {
            String user = etUsername.getText().toString();
            String pass = etPassword.getText().toString();

            if (user.isEmpty() || pass.isEmpty()) {
                Toast.makeText(this, "Fill all fields", Toast.LENGTH_SHORT).show();
            } else {
                Boolean check = db.checkUser(user, pass);
                if (check)
                    Toast.makeText(this, "Login Successful", Toast.LENGTH_SHORT).show();
                else
                    Toast.makeText(this, "Invalid Credentials", Toast.LENGTH_SHORT).show();
            }
        });
    }
}
```

DB

```
package com.example.dblogin;


import android.content.ContentValues;
import android.content.Context;
import android.database.Cursor;
import android.database.sqlite.SQLiteDatabase;
import android.database.sqlite.SQLiteOpenHelper;

public class DBHelper extends SQLiteOpenHelper {

    public static final String DBNAME = "Login.db";

    public DBHelper(Context context) {
        super(context, DBNAME, null, 1);
    }

    @Override
    public void onCreate(SQLiteDatabase db) {
        db.execSQL("create Table users(username TEXT primary key, password TEXT)");
    }

    @Override
    public void onUpgrade(SQLiteDatabase db, int oldVersion, int newVersion) {
        db.execSQL("drop Table if exists users");
        onCreate(db);
    }

    public Boolean insertData(String username, String password) {
        SQLiteDatabase db = this.getWritableDatabase();
        ContentValues values = new ContentValues();

        values.put("username", username);
```

```
        values.put("password", password);

        long result = db.insert("users", null, values);
        return result != -1;
    }

    public Boolean checkUser(String username, String password) {
        SQLiteDatabase db = this.getWritableDatabase();
        Cursor cursor = db.rawQuery(
                "Select * from users where username=? and password=?",
                new String[]{username, password});

        return cursor.getCount() > 0;
    }
}
```

13) LAYOUTS

XML

```
<?xml version="1.0" encoding="utf-8"?>
<ScrollView xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent">

    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:orientation="vertical"
        android:padding="16dp">

        <EditText
            android:id="@+id/u"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:hint="Username"/>

        <EditText
            android:id="@+id/p"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:hint="Password"
            android:inputType="textPassword"/>

        <EditText
            android:id="@+id/a"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:hint="Address"/>
```

```
<RadioGroup
   android:id="@+id/g"
   android:layout_width="wrap_content"
   android:layout_height="wrap_content"
   android:orientation="horizontal">

   <RadioButton
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:text="Male"/>

   <RadioButton
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:text="Female"/>
</RadioGroup>

<EditText
   android:id="@+id/age"
   android:layout_width="match_parent"
   android:layout_height="wrap_content"
   android:hint="Age"
   android:inputType="number"/>

<DatePicker
   android:id="@+id/dp"
   android:layout_width="wrap_content"
   android:layout_height="wrap_content"/>

<Spinner
   android:id="@+id/sp"
   android:layout_width="match_parent"
   android:layout_height="wrap_content"/>

<Button
   android:id="@+id/btn"
   android:layout_width="match_parent"
   android:layout_height="wrap_content"
   android:text="Submit"/>

<TextView
   android:layout_width="wrap_content"
   android:layout_height="wrap_content"
   android:text="Output (Linear Layout):"
   android:textStyle="bold"/>

<TextView
   android:id="@+id/l"
   android:layout_width="match_parent"
   android:layout_height="wrap_content"/>

<TextView
```

```
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Output (Relative Layout):"
        android:textStyle="bold"/>

    <RelativeLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content">

        <TextView
            android:id="@+id/r"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"/>
    </RelativeLayout>

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Output (Table Layout):"
        android:textStyle="bold"/>

    <TableLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content">

        <TableRow>
            <TextView
                android:layout_width="wrap_content"
                android:layout_height="wrap_content"
                android:text="Data:"/>

            <TextView
                android:id="@+id/t"
                android:layout_width="wrap_content"
                android:layout_height="wrap_content"/>
        </TableRow>

    </TableLayout>

  </LinearLayout>
</ScrollView>
```

JAVA

```
package com.example.layout;

import android.os.Bundle;
import android.widget.*;
import androidx.appcompat.app.AppCompatActivity;
```

```
public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle b) {
        super.onCreate(b);
        setContentView(R.layout.activity_main);

        EditText u = findViewById(R.id.u);
        EditText p = findViewById(R.id.p);
        EditText a = findViewById(R.id.a);
        EditText age = findViewById(R.id.age);
        RadioGroup g = findViewById(R.id.g);
        DatePicker dp = findViewById(R.id.dp);
        Spinner sp = findViewById(R.id.sp);
        Button btn = findViewById(R.id.btn);

        TextView l = findViewById(R.id.l);
        TextView r = findViewById(R.id.r);
        TextView t = findViewById(R.id.t);

        sp.setAdapter(new ArrayAdapter<>(this,
                android.R.layout.simple_list_item_1,
                new String[]{"Maharashtra","Gujarat","Delhi"}));

        btn.setOnClickListener(v -> {

            RadioButton rb = findViewById(g.getCheckedRadioButtonId());

            String data =
                    "Username: " + u.getText() +
                            "\nPassword: " + p.getText() +
                            "\nAddress: " + a.getText() +
                            "\nGender: " + rb.getText() +
                            "\nAge: " + age.getText() +
                            "\nDOB: " + dp.getDayOfMonth()+"/"+(dp.getMonth()+1)+"/"+dp.getYear() +
                            "\nState: " + sp.getSelectedItem();

            l.setText(data);
            r.setText(data);
            t.setText(data);
        });
    }
}
```

14 LISTVIEW

XML

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="16dp">

    <EditText
        android:id="@+id/etItem"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter new item"/>

    <Button
        android:id="@+id/btnAdd"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Add Item"/>

    <ListView
        android:id="@+id/listView"
        android:layout_width="match_parent"
        android:layout_height="200dp"/>

    <TextView
        android:id="@+id/tvDetail"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Item Details Appear Here"
        android:textSize="16sp"
        android:padding="10dp"/>

</LinearLayout>
```

JAVA

```
package com.example.listview;

import android.os.Bundle;
import android.widget.*;
import androidx.appcompat.app.AppCompatActivity;
import java.util.ArrayList;

public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle b) {
```

```
        super.onCreate(b);
        setContentView(R.layout.activity_main);

        EditText et = findViewById(R.id.etItem);
        Button add = findViewById(R.id.btnAdd);
        ListView lv = findViewById(R.id.listView);
        TextView tv = findViewById(R.id.tvDetail);

        ArrayList<String> list = new ArrayList<>();
        list.add("Java");
        list.add("Android");
        list.add("Python");

        ArrayAdapter<String> adapter =
                new ArrayAdapter<>(this, android.R.layout.simple_list_item_1, list);
        lv.setAdapter(adapter);

        add.setOnClickListener(v -> {
            list.add(et.getText().toString());
            adapter.notifyDataSetChanged();
            et.setText("");
        });

        lv.setOnItemClickListener((p, v, pos, id) ->
                tv.setText("Selected Item:\n" + list.get(pos)));
    }
}
```